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Brief paper

Enhanced safe control for arbitrary relative degree: A generalized discrete-time CBF approach ☆

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ABSTRACT

Control Barrier Functions (CBFs) are widely used for ensuring safety in control systems and can be implemented on real-world systems via Approximate Sampled-Data Systems (ASDSs). However, the relative degree may be altered during the time-discretization process when generating corresponding ASDSs. Additionally, since the relative degree is a local property, it may vary across the state space. These two facts introduce significant challenges in designing safe controllers. In this paper, we propose a novel approach termed the Generalized Discrete-Time Control Barrier Function (GD-CBF), which provides safety guarantees for ASDSs under certain conditions, regardless of whether the relative degree is constant or variable. A key feature of the GD-CBF framework is the introduction of a safety predictive horizon, which enables improved safety performance and endows the method with predictive capabilities. In addition, it is proved that the safety of the original continuous-time system can be guaranteed in the sense that its state remains within a bounded deviation from the safe set, provided the sampling period is less than a threshold. We also explicitly characterize the dependence of this bound on the sampling period and the properties of the continuous-time system.

1. Introduction

Control Barrier Functions (CBFs) are a powerful extension of classical barrier certificates (Prajna & Jadbabaie, 2004), enabling the direct design of safety-critical controllers with strong theoretical guarantees (Clark, 2021; Jankovic, 2018). CBFs are often combined with control Lyapunov functions to ensure both safety and stability (Romdlony & Jayawardhana, 2016), and they offer lower computational overhead compared to other popular methods such as theorem proving (Mitsch et al., 2013) and reachability analysis (Pek et al., 2020). They have also been integrated with Model Predictive Control (MPC) to jointly achieve system safety and optimal tracking performance (Zeng et al., 2021). In practice, CBFs have been widely adopted in the design of safe controllers for real-world robotic systems (Li et al., 2023; Zeng et al., 2021). In these applications, continuous-time system models are typically discretized to meet the requirements of digital controllers operating in a sampled-data setting, prompting further research into CBFs for sampled-data systems. Specifically, CBFs have been extended to sampled-data frameworks under the zero-order hold (ZOH) assumption, ensuring safety even when the control input is updated only at discrete time instants (Breeden et al., 2021).

For a continuous-time system, the relative degree is the minimum number of times the output needs to be differentiated with respect to time so that its derivative explicitly depends on the input (Sira-Ramírez, 1989). It plays a critical role in determining the structure of the CBF controller, with higher relative degrees posing greater design challenges. However, the relative degree may change during the time-discretization of a continuous-time system. As a result, it is important to investigate how to design discrete-time CBFs that account for changes in relative degree arising from the discretization process.

In particular, the study of relative degree in continuous-time systems typically focuses on three cases: (i) constant relative degree of 1 (Ames et al., 2016), (ii) constant but higher relative degree (Xiao & Belta, 2021), and (iii) variable relative degree across the state space (Brunke et al., 2024). By leveraging numerical methods (such as the Euler method, the Runge–Kutta method, and Taylor expansion), the time discretization of the original continuous-time system during the sampling interval yields an Approximate Sampled-Data System (ASDS) under the

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zero-order hold assumption (Taylor et al., 2022). For systems with constant relative degree 1, it has been shown that sampled-data CBFs can ensure the safety of ASDSs (Taylor et al., 2022). Unfortunately, the relative degree of ASDS usually differs from the original continuous-time system in the second case (Kazantzis & Kravaris, 1997). This phenomenon is usually ignored in existing works designing CBF controllers for ASDSs, where the relative degree of the original continuous-time system is utilized (Jin et al., 2023; Qiu et al., 2024). How this type of simplification impacts the safety guarantee remains unknown. To the best of our knowledge, the explicit relationship between the change of the relative degree induced by discretization and safety performance has yet to be considered. Furthermore, the third case occurs because the relative degree is inherently a local property in the state space (Sira-Ramírez, 1989). The typical CBF method in Ames et al. (2016), which requires a constant relative degree assumption, can lead to safety violations in systems with variable relative degrees. Additionally, variable relative degrees may induce other undesirable behaviors in practical applications, such as trajectory chattering (Brunke et al., 2024, 2025). Thus, the third case increases the risks associated with the design of CBFs. Notably, such a case represents the most challenging scenario—not only does the relative degree vary across the state space, but it can also be altered by the discretization process. Overall, across these three cases, the challenge of how to design a unified CBF for continuous-time systems with arbitrary relative degrees via ASDSs remains unresolved.

In this paper, we introduce a novel method termed Generalized Discrete-time Control Barrier Function (GD-CBF) to ensure the safety of ASDSs. In contrast to existing high-order discrete-time CBFs (Xiong et al., 2022), our approach introduces a novel concept termed *safety predictive horizon*, specifically designed to address arbitrary relative degree cases. We provide an explicit condition on the safety predictive horizon, under which the safety of ASDSs can be ensured even if time-discretization changes the relative degree. This condition is further extended to account for cases where the relative degree varies across the state space. Moreover, we provide an allowable range for the sampling period to ensure that the state of the exact sampled-data system remains within a bounded deviation from the safe set. This bound is further extended — based on the system's Lipschitz continuity — to guarantee that the state between sampling instants also remains within a specified deviation from the safe set. For practical implementation, we design a GD-CBF controller that satisfies a specific condition (see Theorem 2) to ensure the safety of ASDSs. Furthermore, we demonstrate that increasing the value of the safety predictive horizon can endow GD-CBF controllers with predictive capabilities. This predictive capability helps prevent safety violations by evaluating the system's safety over several future steps. These findings were all validated through simulations.

The rest of this paper is organized as follows. The preliminaries and problem formulation are presented in Section 2. Section 3 introduces the proposed GD-CBF method, and the corresponding GD-CBF controller is detailed in Section 4. Simulation results are provided in Section 5, followed by conclusions in Section 6.

Notations: Let $\mathcal{X} \subset \mathbb{R}^n$ be a compact set representing the state space. Similarly, let $\mathcal{U} \subset \mathbb{R}^c$ be a compact set representing the control input space. Let $\mathbb{N}$ denote the set of non-negative integers and let $\mathbb{N}_{>0}$ denote the set of positive integers. Let $\lceil \cdot \rceil$ denote the ceiling function. We use $\mathcal{I}(\cdot)$ to denote the identity function and I to denote the identity matrix. Let $\circ$ denote the composition of relations. The notation $\|\cdot\|$ denotes the norm, specifically the Euclidean norm for real vectors and the Frobenius norm for matrices and tensors. The notation $\backslash$ represents the set minus. A continuous function $\alpha : [-a, b) \to [-a, +\infty)$ is said to be an extended class $\mathcal{K}$ function if it is strictly increasing and $\alpha(0) = 0$.

2. Preliminaries and problem formulation

This section revisits the time-discretization process for continuous-time systems and the relative degree of the resulting ASDSs. Following this, the problem formulation is outlined.

2.1. Taylor expansion-based ASDSs

Consider a continuous-time nonlinear system

$$\dot{x} = f(x, u), \tag{1a}$$

$$y = h(x), \tag{1b}$$

where $x \in \mathcal{X} \subset \mathbb{R}^n$ denotes the state. Here, $u \in \mathcal{U} \subset \mathbb{R}^c$ is the control input, and $y \in \mathbb{R}$ represents the scalar system output. It is assumed that $f(x, u) : \mathcal{X} \times \mathcal{U} \to \mathbb{R}^n$ and $h(x) : \mathcal{X} \to \mathbb{R}$ are (sufficiently) differentiable with respect to x.

The time-discretization of the original continuous-time system (1a) is carried out by expanding the state using a Taylor series over the sampling interval, under the assumption of zero-order hold on the input. Specifically, we define the sampling period as T and the discrete time as $t_k = kT$, where $k \in \mathbb{N}$. It is assumed that the control input $u(t)$ remains constant over the sampling interval $[t_k, t_{k+1})$ (zero-order hold assumption) and is denoted as u_k. The state at time t_k is denoted by x_k. To derive the discrete-time model, we integrate both sides of (1a) over the interval $[t_k, t_{k+1})$, as described in Kazantzis and Kravaris (1999). We further sample the output at t_k, denoted as y_k, resulting in an Exact Sampled-Data System (ESDS) given by

$$x_{k+1} = x_k + \sum_{i=1}^{\infty} \mathcal{A}^{[i]}(x_k, u_k)\frac{T^i}{i!}, \quad y_k = h(x_k). \tag{2}$$

where $\mathcal{A}^{[i]}(x_k, u_k)$ is determined recursively by $\mathcal{A}^{[1]}(x, u) = f(x, u)$ and $\mathcal{A}^{[i+1]}(x, u) = \frac{\partial \mathcal{A}^{[i]}(x,u)}{\partial x}\mathcal{A}^{[1]}(x, u)$ with $i \in \mathbb{N}_{>0}$. The representation in (2) captures the exact discrete-time behavior of (1) by retaining all terms of the infinite Taylor expansion.

However, using an infinite series is computationally infeasible in practice. Thus, (2) is typically truncated at order $N \in \mathbb{N}_{>0}$, where $N < \infty$, resulting in an ASDS of (1a) as follows:

$$x_{k+1} = x_k + \sum_{i=1}^{N} \mathcal{A}^{[i]}(x_k, u_k)\frac{T^i}{i!}, \quad y_k = h(x_k). \tag{3}$$

Here, N is referred to as the "truncation order", which indicates the accuracy of the time-discretization process.

For convenience, although the ASDS and ESDS are distinct systems, we use the same notation x_k to denote their states when no further distinction is necessary.

2.2. Relative degree in ASDSs

This subsection reviews the definition of relative degree in both continuous-time and discrete-time systems and illustrates how continuous-time relative degree changes after discretization.

Definition 1 (*Continuous-time Relative Degree*)**.** The system (1) is said to have a relative degree $r_c \in \{1, \ldots, n\}$ at a point $x \in \mathcal{X}$ if the output function $h(x)$ must be differentiated r_c times along the dynamics of (1) before the control input u appears explicitly (Sira-Ramírez, 1989; Xiao & Belta, 2021).

Definition 2 (*Discrete-time Relative Degree Xiong et al., 2022*)**.** The discrete-time system $x_{k+1} = \bar{f}(x_k, u_k), y_k = h(x_k)$ is said to have relative degree r at state $x_k \in \mathcal{X}$ if and only if there exist functions $g_i : \mathcal{X} \to \mathbb{R}$ and $g_r : \mathcal{X} \times \mathcal{U} \to \mathbb{R}$ such that $y_{k+i} = g_i(x_k),\ \forall i = 0, 1, \ldots, r-1$ and $y_{k+r} = g_r(x_k, u_k)$.

Remark 1. For a continuous-time system, the relative degree is the minimum number of times the output needs to be differentiated so that its derivative explicitly depends on u at $x \in \mathcal{X}$. For a general discrete-time system, the relative degree is the minimal number of time steps r such that u_k explicitly affects the output y_{k+r}.

Typically, the time discretization in (2) is implemented numerically, with the function $g_i(\cdot)$ in Definition 2 usually unknown. Therefore, the relative degree r cannot be determined directly. To address this issue, most practitioners commonly use the relative degree of the original continuous-time system in (1) as a substitute for that of the ASDS (Jin et al., 2023; Qiu et al., 2024). However, this approach can be problematic, as the relative degree of the ASDS may differ from that of the original system, potentially compromising safety guarantees. In fact, Kazantzis and Kravaris (1997) demonstrates that such a discrepancy can occur. This result is formally stated in the following lemma.

Lemma 1 (*Relative Degree in ASDSs Kazantzis & Kravaris, 1997*). *Let r_c be the relative degree of the continuous-time system (1) at a state $x \in \mathcal{X}$. Consider the ASDS (3), obtained by time-discretizing (1) using a Taylor series expansion truncated at order N. Then, the relative degree r of the ASDS at x satisfies $r = \lceil \frac{r_c}{N} \rceil$, except possibly for a countable set of values of the sampling interval T.*

Lemma 1 implies that the time-discretization process changes the relative degree depending on the truncation order N. Consequently, the safe control algorithm must account for this change, which motivates the current work.

2.3. Problem formulation

Given a continuous-time system (1), its corresponding ASDS (3) is obtained by a truncated Taylor expansion. To formulate the problem, we begin by defining two notions of safety.

Definition 3 (*Discrete-time Safety*). A trajectory $\{x_k\}_{k\in\mathbb{N}}$ is said to be discrete-time safe if x_k remains within the set

$$\mathcal{S} = \{x \in \mathcal{X} \mid h(x) \geq 0\} \tag{4}$$

for all $k \in \mathbb{N}$, where $h(x)$ is the output function defined in (1) and $\mathcal{S}$ is referred to as the safe set.

Definition 4 (*Continuous-time Safety*). A trajectory $\{x(t)\}_{t\geq 0}$ is said to be *continuous-time safe* if, for a given $\Delta > 0$, its state $x(t)$ remains within the set $\mathcal{S}_{-\Delta} = \{x \in \mathcal{X} \mid h(x) \geq -\Delta\}$ for all $t \geq 0$, where $h(x)$ is the output function defined in (1).

Remark 2. Note that $\mathcal{S} \subseteq \mathcal{S}_{-\Delta}$ and the set $\mathcal{S}_{-\Delta} \setminus \mathcal{S} = \{x \in \mathcal{X} \mid h(x) \in [-\Delta, 0)\}$ can be interpreted as a buffer zone. This buffer zone represents the region slightly beyond the strict safety boundary defined by $\mathcal{S}$, with the parameter Δ determining the width of this zone. Moreover, since $\mathcal{X}$ is compact and $h(\cdot)$ is continuous, both $\mathcal{S}$ and $\mathcal{S}_{-\Delta}$ are also compact sets.

In this article, although both continuous-time and discrete-time safety are defined in terms of system trajectories, we also refer to a system as safe if and only if all of its trajectories (i.e., solutions) are safe — either in the continuous-time or discrete-time sense — under the specified initial conditions.

Safe controllers designed based on the ASDS (3) are fundamentally influenced by the relative degree of (3). As stated in Lemma 1, the relative degree of (3) is determined by both the truncation order N and the original relative degree $r_c \in \{1, \ldots, n\}$, which makes them important factors to consider in controller design. However, the relative degree is a local property and can vary across the state space $\mathcal{X}$ (Brunke et al., 2024). Thus, more generally, the relative degree of (1) should be redefined in a state-dependent form, i.e., $r_c := \mathcal{R}(x)$, where $\mathcal{R} : \mathcal{X} \to \{1, \ldots, n\}$. This formulation introduces significantly greater challenges for safe controller design.

Based on the previous analysis, this controller should take N and $\mathcal{R}(x)$ into consideration. Accordingly, we define the controller as

$$u_k = \pi(x_k, \mathcal{R}(x), N), \ \forall x \in \mathcal{X}. \tag{5}$$

Our objective is to design the controller in (5) such that under certain initial conditions, (i) all trajectories generated by (3) are discrete-time safe; and (ii) all trajectories generated by (1) are continuous-time safe.

3. Generalized discrete-time control barrier function

This section first proposes the GD-CBF and a condition that ensures the discrete-time safety of the ASDS regardless of whether its relative degree is constant or state-dependent. Moreover, we provide an allowable range for the sampling period to ensure the safety performance of the original system does not degrade significantly compared to that of the ASDS.

3.1. Formulation of GD-CBF

To construct the GD-CBF, we define a series of functions that extend the corresponding functions presented in Xiong et al. (2022). Assuming the relative degree of ASDS (3) at state x_k is r, we first define a series of functions based on output $h(x_k)$ as follows[1]:

$$\phi_0(x_k) := h(x_k), \tag{6a}$$
$$\phi_1(x_k) := \Delta\phi_0(x_k) + \alpha_0(\phi_0(x_k)), \tag{6b}$$
$$\vdots$$
$$\phi_r\left(\tau_k^0\right) := \Delta\phi_{r-1}(x_k) + \alpha_{r-1}(\phi_{r-1}(x_k)), \tag{6c}$$
$$\phi_{r+1}\left(\tau_k^1\right) := \Delta\phi_r\left(\tau_k^0\right) + \alpha_r\left(\phi_r\left(\tau_k^0\right)\right), \tag{6d}$$
$$\vdots$$
$$\phi_m(\tau_k^d) := \Delta\phi_{m-1}(\tau_k^{d-1}) + \alpha_{m-1}(\phi_{m-1}(\tau_k^{d-1})), \tag{6e}$$

where the equations from (6d) to (6e) are newly introduced compared to those in Xiong et al. (2022). Here, $m \in \mathbb{N}_{>0}$ is defined as the "safety predictive horizon", representing the maximum number of future steps of anticipated ASDS outputs considered when enforcing safe constraints. We define d as $d := m - r$. From (6a) to (6c), $\Delta\phi_i(x_k)$ denotes $\phi_i(x_{k+1}) - \phi_i(x_k)$, where $i = 0, 1, \ldots, r-1$. From (6d) to (6e), $\Delta\phi_{r+j}(\tau_k^j)$ represents $\phi_{r+j}(\tau_{k+1}^j) - \phi_{r+j}(\tau_k^j)$, where

$$\tau_k^j = \{x_k, u_k, u_{k+1}, \ldots, u_{k+j}\}$$

and $j = 0, 1, \ldots, d-1$. Additionally, $\alpha_l(\cdot)$, $l = 0, 1, \ldots, m-1$ are extended class $\mathcal{K}$ functions that satisfies $\alpha_l(z) \leq z$ for all $z > 0$.

Remark 3. The equations from (6d) to (6e) exist only when $m > r$. If $m < r$, $\phi_m(\tau_k^d)$ is redefined as $\phi_m(x_k)$, indicating that $\phi_m(x_k)$ is not affected by any control input. The cases where $m < r$ are commonly observed in systems with variable relative degrees. Interestingly, if we choose all $\alpha_l(\cdot) = \mathcal{I}(\cdot)$ in (6) for $l = 0, \ldots, m-1$, then $\phi_m(\tau_k^d)$ reduces exactly to $h(x_{k+m})$.

[1] We use r to simplify the definition and analysis of the GD-CBFs, where the full, rigorous expression is given by $\lceil \frac{r_c}{N} \rceil$.

The GD-CBF is defined based on (6), which is presented as follows.

Definition 5 (*GD-CBF*)**.** For the ASDS (3) and a selected safety predictive horizon m, given $\phi_i(\cdot)$, $i \in \{0, 1, \dots, m\}$ defined by (6) and $\mathcal{S}$ defined by (4), the continuous function $h : \mathbb{R}^n \to \mathbb{R}$ is a GD-CBF if there exist control inputs $u_k, \dots, u_{k+d} \in \mathcal{U}$ that satisfy

$$\phi_m\left(\tau_k^d\right) \geq 0 \tag{7}$$

for all $x_k \in \mathcal{S}$.

Remark 4. The concept of safety predictive horizon m is inspired by the predictive horizon used in MPC. Given a GD-CBF, m indicates how many anticipated outputs $h(x_{k+i})$, $i \in \mathbb{N}_{>0}$ are considered in the safe constraint (7). The safety predictive horizon is a key parameter in determining the formulation in (7), and therefore plays a direct role in the design of the safe controller.

Remark 5 (*Relationship Between the GD-CBF and High-Order Discrete-Time CBF in Xiong et al., 2022*)**.** This GD-CBF method can generally be applied to more general discrete-time systems when starting directly from the form in (3). In view of this, GD-CBF can be considered an extension of the high-order discrete-time CBF proposed in Xiong et al. (2022). Specifically, our GD-CBF extracts the concept of the safety predictive horizon m, allowing $m \neq r$, unlike the stricter requirement $m \equiv r$ in Xiong et al. (2022). The safety predictive horizon provides greater flexibility in the design of safe controllers across various scenarios.

The following two subsections provide a detailed analysis of discrete-time safety for the ASDS and continuous-time safety for the original system (1).

3.2. Discrete-time safety analysis

This subsection derives a condition on the safety predictive horizon m to ensure the discrete-time safety of ASDSs, initially under the assumption that the relative degree of the continuous-time system (1) is constant. We then extend this condition to handle cases where the relative degree varies across the state space.

We begin by assuming that the relative degree of (1) is constant and denoted by r_c. Under this assumption, the condition for m is derived based on the following proposition.

Proposition 1. *Given the functions $\phi_{r+j}(\tau_k^j)$, $j = 0, \dots, d-1$ defined in (6), if the control inputs $u_k, \dots, u_{k+j}$ satisfy*

$$\Delta\phi_{r+j}\left(\tau_k^j\right) + \alpha_{r+j}\left(\phi_{r+j}\left(\tau_k^j\right)\right) \geq 0 \tag{8}$$

for all $k \in \mathbb{N}$ and $\phi_{r+j}(\tau_0^j) \geq 0$, then $\phi_{r+j}(\tau_k^j) \geq 0$ holds for all $k \in \mathbb{N}$.

Proof. The proof follows directly from Ahmadi et al. (2019, Theorem 1) and is therefore omitted.

Theorem 1. *Consider an original continuous-time system (1) with a global relative degree r_c. Furthermore, consider that it is time-discretized with a truncated order N, yielding an ASDS (3). Then, given the initial conditions*

$$\phi_i(x_0) \geq 0,\ i = 0, 1, \dots, r-1, \tag{9a}$$

$$\phi_{r+j}(\tau_0^j) \geq 0,\ j = 0, 1, \dots, d-1, \tag{9b}$$

if $h(x_k)$ is a GD-CBF with safety predictive horizon m satisfying

$$m \geq r = \left\lceil \frac{r_c}{N} \right\rceil, \tag{10}$$

then any discrete-time control inputs $u_k, \dots, u_{k+d}$ render (3) discrete-time safe, except possibly for a countable set of values of sampling period T.

Proof. According to Lemma 1, the relative degree of ASDS (3) is $r = \left\lceil \frac{r_c}{N} \right\rceil$ with its truncation order N except possibly for a countable set of values of T. For $m = r$, it has been proven that $\phi_{r-1}(x_k) \geq 0$ holds if (9a) and $\phi_r(\tau_k^0) \geq 0$ are satisfied (see Ahmadi et al. (2019, Theorem 1)). We extend this result to the condition $m \geq r$ as follows.

For $m > r$, according to Proposition 1, the discrete-time control inputs $u_k, \dots, u_{k+d}$ that satisfy (7) and $\phi_{r+d-1}(\tau_0^{d-1}) \geq 0$ can ensure $\phi_{r+d-1}(\tau_k^{d-1}) \geq 0$ for $k \in \mathbb{N}$. Then, $u_k, \dots, u_{k+d}$ can also ensure $\phi_{r+d-2}(\tau_k^{d-2}) \geq 0$ if $\phi_{r+d-2}(\tau_0^{d-2}) \geq 0$. Iteratively, these discrete-time control inputs will ensure $\phi_r(\tau_k^0) \geq 0$ if initial conditions (9b) hold. Thus, $m > r$ can also render (3) discrete-time safe according to Ahmadi et al. (2019, Theorem 1).

Finally, the satisfaction of initial conditions (9) and constraint (7) can render (3) discrete-time safe if $m \geq \left\lceil \frac{r_c}{N} \right\rceil$, except possibly for a countable set of values of T. This completes the proof.

Theorem 1 provides a lower bound for selecting the safety predictive horizon m to guarantee the discrete-time safety of ASDS. A larger m incorporates more future outputs y_{k+i} into the constraint (7), potentially enhancing safety. However, Theorem 1 applies only when r_c is constant, making it insufficient for scenarios with variable relative degree. To address this limitation, we extend Theorem 1 and provide a condition on the safety predictive horizon m to ensure the safety of (3) when r_c varies across the state space.

Theorem 2. *Consider the continuous-time systems in (1), and let (3) denote their ASDSs obtained via time discretization with truncation order N. Under the initial conditions (9) and assuming the controller enforces the constraint (7), then: (i) the ASDS is discrete-time safe if $m \geq \left\lceil \frac{\max_x \mathcal{R}(x)}{N} \right\rceil$, except possibly for a countable set of sampling periods T; (ii) there exists an ASDS of the form (3) such that if $m < \left\lceil \frac{\max_x \mathcal{R}(x)}{N} \right\rceil$, then it is not discrete-time safe.*

Proof. (i) If $m \geq \left\lceil \frac{\max_x \mathcal{R}(x)}{N} \right\rceil$, then it follows that $m \geq \left\lceil \frac{\mathcal{R}(x)}{N} \right\rceil$ for all $x \in \mathcal{X}$. Under this condition, when the controller enforces the GD-CBF constraint (7), the discrete-time safety of all trajectories of (3) can be directly ensured by Theorem 1.

(ii) Consider the following illustrative continuous-time system

$$\begin{cases} \dot{x}^{(1)} = f(x^{(2)}) + u, \\ \dot{x}^{(2)} = x^{(1)} + g(x^{(2)})u, \\ \quad y = h(x) = g(x^{(2)}), \end{cases} \tag{11}$$

where $x = [x^{(1)}, x^{(2)}]^\top \in \mathbb{R}^2$, and $g : \mathbb{R} \to \mathbb{R}$ is a continuous function. Assume that unsafe states exist, that is, there exists a nonempty $\bar{\mathcal{X}} \subset \mathcal{X} \subset \mathbb{R}^2$ such that $h(x) < 0$ if $x \in \bar{\mathcal{X}}$. The existence of $x, y \in \mathcal{X}$ such that $h(x) < 0$ and $h(y) \geq 0$, together with the continuity of $g(\cdot)$, implies that the zero-level set $\mathcal{B} := \{x^{(2)} \mid g(x^{(2)}) = 0\}$ is nonempty. According to Definition 1, the relative degree of (11) is given by

$$\mathcal{R}(x) = \begin{cases} 1, & x^{(2)} \notin \mathcal{B}, \\ 2, & x^{(2)} \in \mathcal{B}. \end{cases}$$

With truncation order $N = 1$, the corresponding ASDS of (11) is

$$\begin{cases} x_{k+1}^{(1)} = x_k^{(1)} + T(f(x_k^{(2)}) + u_k), \\ x_{k+1}^{(2)} = x_k^{(2)} + T(x_k^{(1)} + g(x_k^{(2)})u_k), \\ \quad y_k = h(x_k) = g(x_k^{(2)}), \end{cases} \tag{12}$$

where $x_k = [x_k^{(1)}, x_k^{(2)}]^\top$. Since $\lceil \frac{\max_x \mathcal{R}(x)}{N} \rceil = 2$, the condition $m < \lceil \frac{\max_x \mathcal{R}(x)}{N} \rceil$ corresponds to selecting $m = 1$.

Now consider a GD-CBF controller with $m = 1$ for which the initial condition of (11) is $g(x_0^{(2)}) \geq 0$. Choose an arbitrary $x_0^{(2)}$ such that $g(x_0^{(2)}) > 0$, yielding $\mathcal{R}(x_0) = 1$. Since the controller enforces the constraint (7), u_0 is designed to guarantee that $h(x_1) \geq 0$. Let u_0 be selected such that $x_1^{(2)} \in \mathcal{B}$, which obviously satisfies (7). Then,

$$x_2^{(2)} = x_1^{(2)} + T(x_0^{(1)} + Tf(x_0^{(2)}) + u_0),$$

which is independent of u_1 since $g(x_1^{(2)}) = 0$. Moreover, note that there exists Δ such that $g(x_1^{(2)} + \Delta) < 0$ since $\bar{\mathcal{X}}$ is nonempty and $g(\cdot)$ is continuous. Because $x_0^{(1)}$ is not constrained by the safety requirement, choosing $x_0^{(1)} = \Delta/T - Tf(x_0^{(2)}) - u_0$, it then follows that $g(x_2^{(2)}) < 0$. Thus, (12) is not discrete-time safe. This illustrates that a controller without a sufficient safety predictive horizon cannot guarantee discrete-time safety, and completes the proof.

Theorem 2 (*ii*) proves the existence of ASDS that is not discrete-time safe if $m < \lceil \frac{\max_x \mathcal{R}(x)}{N} \rceil$. Intuitively, without sufficient predictive capability when $m = 1$, x_k is likely to evolve uncontrollably into the unsafe region from the boundary of the safe set where the relative degree is 2. Beyond the case described in Theorem 2, unsafe state trajectories can also be caused by unrestricted control inputs, as illustrated in Section 5.2.

Corollary 1. *If truncation order N satisfies that $N \geq \max_x \mathcal{R}(x)$, then condition (7) with the safety predictive horizon of $m = 1$ can render (3) discrete-time safe.*

Theorem 2 and Corollary 1 indicate that increasing either the safety predictive horizon m or the truncation order N can enhance the safety of ASDSs. More importantly, since $\mathcal{R}(x)$ encompasses all cases of relative degree, Theorem 2 is applicable for ASDSs with arbitrary relative degree cases.

3.3. Continuous-time safety analysis

In general, safety cannot be fully ensured during the interval between sampling instants t_k and t_{k+1} without additional assumptions. Nonetheless, this subsection ensures that the safety performance of the original system (1) does not degrade significantly compared to that of the ASDS.

Our second objective is to ensure that the system (1) remains within the set $\mathcal{S}_{-\Delta}$ under certain initial conditions, meaning it either stays in the original safe set $\mathcal{S}$ or within the buffer zone $\{x \in \mathcal{X} \mid h(x) \in [-\Delta, 0)\}$. This objective is achieved in two steps: (i) ensure that (1) satisfies $h(x_k)$ for all $k \in \mathbb{N}$ under certain initial conditions; (ii) ensure that (1) satisfies $h(x(t)) \geq -\Delta$ for all $t > 0$ under certain initial conditions. Our analysis proceeds under the following assumptions.

Assumption 1. The original system (1) satisfies:

- $\exists f_{\max} > 0$ such that $\|f(x, u)\| \leq f_{\max}$, $\forall x \in \mathcal{X},\ u \in \mathcal{U}$.
- $\exists \mathbb{L}_f^i > 0$ for $i \in \{1, \dots, N+1\}$ such that $\|\frac{\partial^i f(x,u)}{\partial x^i}\| \leq \mathbb{L}_f^i$, $\forall x \in \mathcal{X},\ u \in \mathcal{U}$.

Assumption 2. $h(x)$ is Lipschitz continuous with Lipschitz constant $\mathbb{L}_h > 0$.

Lemma 2. *Under Assumption 1, for any integer $n \geq 1$, the norm of $\mathcal{A}^{[n+1]}(x, u)$ satisfies:*

$$\|\mathcal{A}^{[n+1]}(x, u)\| \leq Mn!, \tag{13}$$

where $M = \max_{\lambda \vdash n}(\prod_{k=1}^{\ell} \mathbb{L}_f^{\lambda_k})(f_{max})^{n+1-\ell}$. Here, $\lambda \vdash n$ denotes all integer partitions $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_\ell)$ with partition length ℓ such that $\sum_{k=1}^{\ell} \lambda_k = n$.

Proof. The proof proceeds in three steps. For brevity, we occasionally denote $f(x, u)$ and $\mathcal{A}^{[i+1]}(x, u)$ by f and $\mathcal{A}^{[i+1]}$, respectively, throughout the proof.

Step 1: This step establishes that $\mathcal{A}^{[n+1]}$ admits the form

$$\mathcal{A}^{[n+1]} = \sum_{\lambda \vdash n} c_\lambda \left(\prod_{k=1}^{\ell} \frac{\partial^{\lambda_k} f}{\partial x^{\lambda_k}} \right) f^{n+1-\ell}, \tag{14}$$

where c_λ denotes the coefficient associated with the partition λ. For $n = 1$, we have $\mathcal{A}^{[2]} = \frac{\partial f}{\partial x} f$, which corresponds to the partition $\lambda = (1)$ with $c_{(1)} = 1$, matching (14). Further, assume (14) holds for $n = i$. Then for $n = i + 1$, it follows that $\mathcal{A}^{[i+2]} = \sum_{\lambda \vdash i} c_\lambda \frac{\partial}{\partial x} \left[\left(\prod_{k=1}^{\ell} \frac{\partial^{\lambda_k} f}{\partial x^{\lambda_k}} \right) f^{i+1-\ell} \right] f$. Applying the product rule to each $\left(\prod_{k=1}^{\ell} \frac{\partial^{\lambda_k} f}{\partial x^{\lambda_k}} \right) f^{i+1-\ell}$, one obtains:

$$\begin{aligned}\mathcal{A}^{[i+2]} = \sum_{\lambda \vdash i} c_\lambda \Bigg[&\sum_{j=1}^{\ell} \left(\prod_{k \neq j} \frac{\partial^{\lambda_k} f}{\partial x^{\lambda_k}} \frac{\partial^{\lambda_j + 1} f}{\partial x^{\lambda_j + 1}} \right) f^{i+2-\ell} \\ &+ \left(\prod_{k=1}^{\ell} \frac{\partial^{\lambda_k} f}{\partial x^{\lambda_k}} \right) (i + 1 - \ell) \frac{\partial f}{\partial x} f^{i+1-\ell} \Bigg].\end{aligned} \tag{15}$$

This yields new partitions $\mu \vdash i + 1$ by either (i) incrementing a component $\lambda_j \rightarrow \lambda_j + 1$ or (ii) appending a unit partition element $\lambda \cup \{1\}$. The corresponding coefficient update is $c_\mu = \sum_{\lambda \rightarrow \mu} c_\lambda + \sum_{\lambda \cup \{1\} = \mu} c_\lambda (i + 1 - \ell)$. Thus, the inductive step holds, and by induction, (14) is valid for all $n \geq 1$.

Step 2: Since the coefficients c_λ are independent of the specific form of $f(x, u)$, to determine $\sum_{\lambda \vdash n} c_\lambda$, we may consider the particular choice $f(x, u) = e^x$. In this context, one has $\mathcal{A}^{[2]} = \frac{d}{dx}(e^x) \cdot e^x = e^{2x}$. Assume $\mathcal{A}^{[i+1]} = i! e^{(i+1)x}$. Then, one obtains $\mathcal{A}^{[i+2]} = (i+1)! e^{(i+2)x}$, which completes the inductive step. Hence, $\mathcal{A}^{[n+1]} = n! e^{(n+1)x}$, implying that $\sum_{\lambda \vdash n} c_\lambda = n!$.

Step 3: Define $\mathcal{F}_{n,\lambda} := \left(\prod_{k=1}^{\ell} \frac{\partial^{\lambda_k} f}{\partial x^{\lambda_k}} \right) f^{n+1-\ell}$. Using the triangle inequality and homogeneity of norms, one has $\| \sum_{\lambda \vdash n} c_\lambda \mathcal{F}_{n,\lambda} \| \leq \sum_{\lambda \vdash n} c_\lambda \|\mathcal{F}_{n,\lambda}\|$. By Assumption 1 and the Cauchy–Schwarz inequality, each term can be bounded as $\|\mathcal{F}_{n,\lambda}\| \leq (\prod_{k=1}^{\ell} \mathbb{L}_f^{\lambda_k}) (f_{\max})^{n+1-\ell}$. Then, combining the coefficient sum from Step 2, it follows that $\sum_{\lambda \vdash n} c_\lambda \|\mathcal{F}_{n,\lambda}\| \leq Mn!$. This completes the proof.

Theorem 3. *Consider the ASDS (3) with truncation order N. Suppose that Assumptions 1 and 2 hold, and that the GD-CBF condition (7) and the initial conditions (9) are satisfied. For a given $\delta > 0$, if the sampling period T satisfies*

$$T \leq \sqrt[N+1]{\frac{(N + 1)\eta}{\mathbb{L}_h M}}, \tag{16}$$

where $\eta := \min\{\delta, -\alpha_0(-\delta)\}$, then $x_k \in \mathcal{S}_{-\delta}$ for all $k \in \mathbb{N}$.

Proof. To distinguish between the ASDS and ESDS states in this proof, let x_{k+1} continue to denote the discrete-time state of the original continuous-time system (1), and let $\hat{x}_{k+1}$ denote the ASDS state. From the ESDS in (2) and ASDS in (3), we can express the ESDS state as $x_{k+1} = \hat{x}_{k+1} + \frac{T^{N+1}}{(N+1)!} \mathcal{A}^{[N+1]}(x(\xi), u_k)$ for some $\xi \in [t_k, t_{k+1})$. According to Lemma 2, the approximate error is therefore bounded by $\|x_{k+1} - \hat{x}_{k+1}\| \leq \frac{MT^{N+1}}{(N+1)}$. Under Assumption 2, one further obtains

$$|h(\hat{x}_{k+1}) - h(x_{k+1})| \leq \frac{\mathbb{L}_h M T^{N+1}}{(N + 1)}. \tag{17}$$

We now consider two cases, based on whether the current state x_k lies inside the safe set $\mathcal{S}$, or in the buffer region $\mathcal{S}_{-\delta} \setminus \mathcal{S}$.

Case 1: For any $x_k \in \mathcal{S}$, by Theorem 1, we have $h(\hat{x}_{k+1}) \geq 0$. Combining this with (17), we obtain $h(x_{k+1}) \geq -\delta$ whenever $\frac{\mathbb{L}_h M T^{N+1}}{N+1} \leq \delta$.

Case 2: For any $x_k \in \mathcal{S}_{-\delta}/\mathcal{S}$, we have $h(x_k) \in [-\delta, 0)$. From the GD-CBF condition (7), it follows that $h(\hat{x}_{k+1}) \geq -\delta - \alpha_0(-\delta)$. According to (17), we have $h(x_{k+1}) \geq h(\hat{x}_{k+1}) - \frac{\mathbb{L}_h M T^{N+1}}{(N+1)}$. Then $h(x_{k+1}) \geq -\delta$ holds whenever $\frac{\mathbb{L}_h M T^{N+1}}{(N+1)} \leq -\alpha_0(-\delta)$.

To ensure $x_{k+1} \in \mathcal{S}_{-\delta}$ in both cases, it suffices to choose the sampling period T such that $\frac{\mathbb{L}_h M T^{N+1}}{(N+1)} \leq \min\{\delta, -\alpha_0(-\delta)\}$. Solving for T completes the proof.

Remark 6. Theorem 3 imposes an implicit constraint on the truncation order N if the sampling period is fixed, i.e., $\frac{(N+1)\eta}{\mathbb{L}_h M} \geq T^{N+1}$. Such a constraint can be used to verify if the truncation order N is appropriate.

Theorem 3 guarantees that $h(x_k) \geq -\delta$ holds for the original system based on Theorem 2. Additionally, Theorem 3 provides practical guidance on selecting an appropriate sampling period to ensure safety.

The remaining step to establish continuous-time safety of the original system (1) is completed by the following theorem.

Theorem 4 (*Continuous-time Safety*)**.** *Consider the original continuous-time system (1) and its associated ASDS (3), operating under the sampling period condition specified in Theorem 3. Suppose the GD-CBF constraints (7) and the initial conditions (9) are satisfied. Then, the original system (1) is continuous-time safe with respect to the expanded safe set $\mathcal{S}_{-\Delta}$, where $\Delta := \delta + \mathbb{L}_h f_{\max} \frac{T}{2}$.*

Proof. We analyze the state trajectory $x(t)$ over each sampling interval $[t_k, t_{k+1})$. Under the zero-order hold assumption, the control input remains constant at u_k, and the state evolves as $x(t) = x_k + \int_{t_k}^{t} f(x(\tau), u_k)d\tau$. This implies the bound $\|x(t) - x_k\| \leq f_{\max}(t - t_k)$. From Theorem 3, we know that $h(x_k) \geq -\delta$ for all $k \geq 0$. Then, for a fixed $k \in \mathbb{N}$ and any $t \in [t_k, t_{k+1})$, one has

$$h(x(t)) \geq -\delta - \mathbb{L}_h f_{\max}(t - t_k). \tag{18}$$

Similarly, using $h(x_{k+1}) \geq -\delta$, an alternative lower bound for $h(x(t))$ is given by:

$$h(x(t)) \geq -\delta - \mathbb{L}_h f_{\max}(t_{k+1} - t). \tag{19}$$

By combining inequalities (18) and (19), which are both valid over the interval $t \in [t_k, t_{k+1})$, a unified lower bound for $h(x(t))$ is $\max\{-\delta - \mathbb{L}_h f_{\max}(t - t_k),\ -\delta - \mathbb{L}_h f_{\max}(t_{k+1} - t)\}$. The minimum of $h(x(t))$ — i.e., the worst-case safety violation — occurs at the midpoint $t = t_k + \frac{T}{2}$, yielding $h(x(t)) \geq -\Delta$. Thus, $x(t) \in \mathcal{S}_{-\Delta}$ holds for all $t \geq 0$.

Theorem 4 provides a basis for practical safety enhancements such as obstacle expansion. For example, if $\Delta = 0.1$ is chosen for continuous-time safety, one can simply shift the obstacle boundary so that a previous value $h(x) = 0.1$ is treated as $h(x) = 0$.

4. Controller design via GD-CBF

This section provides the GD-CBF controller based on the condition stated in Theorem 2. Initially, we examine the predictive capability of the proposed GD-CBF. We define *predictive capability* as the property wherein the control input u_k that satisfies the safe constraint (7) is influenced by the value of anticipated future outputs $h(x_{k+i})$, $i \in \mathbb{N}_{>0}$. The predictive capability indicates that the behavior of ASDS (3) adapts in response to anticipated future safe conditions.

Algorithm 1 GD-CBF Controller

```
while true do
    Initialize ψ_0[i] ← h(x_{k+i}) for i = 0, ..., m
    for i = 1 to m do
        for j = 0 to m − i do
            ψ_i[j] ← ψ_{i−1}[j + 1] − ψ_{i−1}[j] + α_{i−1}(ψ_{i−1}[j])
        end for
    end for
    φ_m(τ_k^d) ← ψ_m[0]; u_k ← solution of (20); k ← k + 1
end while
```

However, not all GD-CBFs exhibit predictive capability; rather, it necessitates two conditions: (1) $\phi_m(\tau_k^d)$ in (7) must include several anticipated future outputs $h(x_{k+i})$, $i \in \mathbb{N}_{>0}$; (2) the control input u_k appears in $\phi_m(\tau_k^d)$. Actually, $\phi_m(\tau_k^d)$ contains $h(x_{k+1}), \ldots, h(x_{k+m})$, as deduced from (6). Hence, the first condition is always satisfied for all GD-CBFs. The second condition holds when $m \geq \lceil \frac{r_{\max}}{N} \rceil$. Otherwise, there always exists an $x_k \in \mathcal{X}$ such that $\phi_m(x_k)$ becomes independent of u_k (see the proof of Theorem 2). Therefore, we conclude that the GD-CBF exhibits predictive capability for all $x_k \in \mathcal{X}$ if $m \geq \lceil \frac{r_{\max}}{N} \rceil$. The predictive capability of GD-CBF can prevent trajectory safety violations by incorporating certain future control inputs within the constraint (7) (Bejarano et al., 2023).

Remark 7. While GD-CBF exhibits predictive capability for large m, there is a trade-off between this capability and the computational burden. Specifically, constructing the GD-CBF constraint in (7) incurs a computational complexity of $\mathcal{O}(Nn^2m^2)$. As a result, increasing m significantly raises the computational load. To balance safety and real-time performance, we advise selecting m close to the minimum required value, given by $\lceil \frac{\max_{x \in \mathcal{X}} \mathcal{R}(x)}{N} \rceil$, unless stricter safety requirements necessitate a larger m.

The predictive capability of GD-CBF can facilitate its integration into the design of a controller similar to MPC. The designed controller, referred to as the GD-CBF controller, not only achieves the task objectives but also ensures the safety of ASDSs. Its formulation is outlined as follows:

$$\min_{u_{k:k+d-1}} \sum_{i=0}^{N_h-1} L\left(x_{k+i}, u_{k+i}\right) + E\left(x_{k+N_h}\right) \tag{20a}$$

$$\text{s.t. } (3), (7), \tag{20b}$$

$$x_{k+i} \in \mathcal{X}, u_{k+i} \in \mathcal{U},\ i = 0, \ldots, N_h - 1, \tag{20c}$$

$$\| u_{k+i} - u_{k+i-1} \| \leq \Delta_u,\ i = 1, \ldots, N_h - 1. \tag{20d}$$

In this controller, N_h is the prediction horizon and Δ_u represents the maximum allowed change in the control inputs. $u_{k:k+d}$ denotes control inputs $u_k, \ldots, u_{k+d}$. The terms $L(x_{k+i}, u_{k+i})$ and $E(x_{k+N_h})$ denote stage cost and terminal cost, respectively. For convenience, they are typically defined as $L(x_{k+i}, u_{k+i}) = \delta(x_{k+i})^T Q \delta(x_{k+i}) + [u_{k+i}]^T R [u_{k+i}]$ and $E(x_{k+N_h}) = x_{k+N_h}^T Q_f x_{k+N_h}$ with $\delta(x_{k+i}) = x_{k+i} - x_{\text{ref}}$, where Q, R, and Q_f are symmetric positive-definite matrices. The GD-CBF controller is computed repeatedly, but only the first control input u_k is implemented at each step. Additionally, the value of m determines the predictive capability of the GD-CBF controller (20). The detailed procedure for the GD-CBF controller is presented in Algorithm 1. In the subsequent simulations, the corresponding ASDSs in (3) are updated, and m in (7) is adjusted when deploying the GD-CBF controllers.

Remark 8. If the functions $\phi_i(\cdot)$, $i \in \{0, 1, \ldots, m\}$ define a valid GD-CBF, there always exists a control input sequence $\{u_k, \ldots, u_{k+N_h-1}\}$ that satisfies both $u_{k+i} \in \mathcal{U},\ i = 0, \ldots, N_h-1$ and $\phi_m\left(\tau_k^d\right) \geq 0$ (see Definition 5). However, if Δ_u is set too small, feasibility may be compromised by another constraint: $\|u_{k+i} - u_{k+i-1}\| \leq \Delta_u$ for $i = 1, \ldots, N_h-1$. To maintain feasibility, either Δ_u must be sufficiently large or the control input space $\mathcal{U}$ must be adequately sized.

Remark 9 (*Comparison with MPC-CBF*). The MPC-CBF controller imposes the following iterative constraints to replace (7):

$$\Delta h\left(x_{k+i}, u_{k+i}\right) \geq -\gamma h\left(x_{k+i}\right), \quad i = 0, \ldots, N_h - 1, \tag{21}$$

where $\Delta h(x_{k+i}, u_{k+i}) = h(x_{k+i+1}) - h(x_{k+i})$ and $\gamma \in (0, 1]$ is a constant parameter. Both controllers exhibit predictive capabilities. However, the MPC-CBF controller faces challenges with systems of high relative degree. Specifically, for (21) with $i = 0, \ldots, r-1$, the term $\Delta h(x_{k+i}, \cdot)$ is independent of the control inputs $u_k, \ldots, u_{k+i}$ (see Definition 2), rendering the first r constraints difficult to satisfy. This often results in significantly increased computational complexity. In contrast, although the GD-CBF controller's optimization problem is typically nonconvex, similar to MPC-CBF, its constraint in (7) consistently depends on the control inputs $u_k, \ldots, u_{k+d}$. Consequently, the GD-CBF controller may achieve reduced computation times for systems with high relative degrees.

5. Case study

This section demonstrates the effectiveness of the proposed GD-CBF controller (20) through two case studies.

5.1. Quadrotor attitude control

This simulation aims to validate Theorems 1–4 and to demonstrate the advantages of the GD-CBF framework compared to existing methods. We consider a quadrotor attitude dynamics model:

$$\dot{R} = R\hat{\Omega}, \tag{22a}$$

$$\dot{\Omega} = J^{-1}\Omega \times J\Omega + J^{-1}u. \tag{22b}$$

where $R \in \mathbb{R}^{3\times3}$ is the rotation matrix from the body-fixed frame to the world frame, representing the quadrotor's attitude. Here, $J \in \mathbb{R}^{3\times3}$ is the inertia matrix and $\Omega \in \mathbb{R}^3$ is the body angular velocity. The symbol $u \in \mathbb{R}^3$ refers to the control input, also known as the total moment in the body-fixed frame. The hat map $\hat{(\cdot)} : \mathbb{R}^3 \to \mathbb{R}^{3\times3}$ satisfies $\hat{a}b = a \times b$ for any $a, b \in \mathbb{R}^3$. The system state is defined as $x = [\mathrm{vec}(R), \Omega]^\top$, consistent with the definition in (1), where $\mathrm{vec}(\cdot)$ denotes matrix vectorization.

The output function is defined as:

$$h(x) = v^\top Rz - \cos(\theta), \tag{23}$$

where v denotes the desired safe direction (typically vertical), and $z := [0, 0, 1]^\top$ represents the body-frame z-axis. The term Rz denotes the quadrotor body's z-axis expressed in world coordinates. Thus, $v^\top Rz$ quantifies the alignment between the body's z-axis and the safe direction v. Setting $v = [0, 0, 1]^\top$ and $\theta = \pi/6$ constrains the quadrotor's tilt to within 30° of vertical. The relative degree of (22) with respect to $h(x)$ is $r_c = 2$. The objective is to ensure $h(x_k) \geq 0,\ \forall k \in \mathbb{N}$ for the ASDS while tracking a reference trajectory.

To verify the sufficiency of condition (10) for ensuring the safety of the ASDS obtained from (22), we test three truncation orders: $N = 1$, 2, and 3. For each order, we vary the safety predictive horizon $m \in \{1, 2, 3, 4\}$, producing 12 scenarios. Each

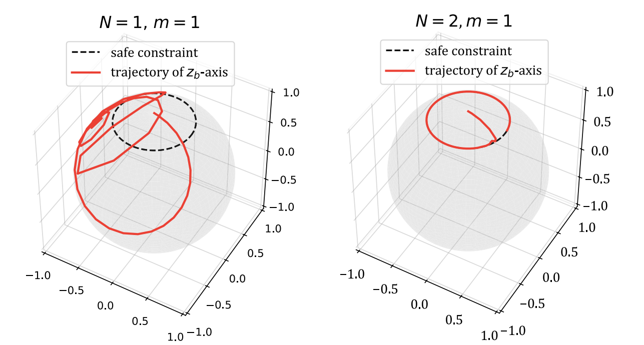

Fig. 1. Trajectories of the quadrotor body's z-axis in the world frame under two representative configurations.

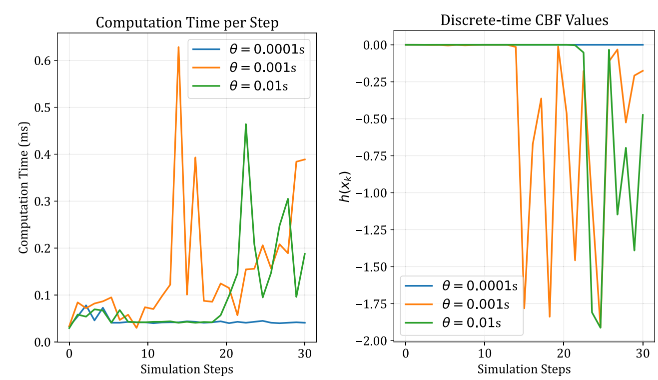

Fig. 2. Computation time and CBF values for different values of θ. Smaller θ values push the safety boundary closer to the critical state where $\mathcal{R}(x) = 3$.

configuration is tested with 10 random initial attitudes within the safe set. The results consistently show that configurations violating $m \geq \lceil r_c/N \rceil$ lead to unsafe cases — for example, the setting $N = m = 1$ yields unsafe trajectories — whereas all configurations satisfying the condition achieve a 100% safety rate. Fig. 1 further illustrates representative cases. These findings confirm that the condition (10) reliably ensures safety for the considered ASDSs.

For states where $R_{[1,3]} = R_{[2,3]} = 0$ and $R_{[3,3]} = 1$ (e.g., $R = I$), the relative degree of the system (22) increases to $r_c = 3$. We refer to such states as "critical states". In the vicinity of these states, Theorem 1 may no longer ensure safety when $m = 2$. To investigate this behavior, we vary the parameter θ in (23) across values $\{0.1, 0.01, 0.001\}$, tightening the safety constraint and pushing the system closer to the critical states. As illustrated in the right subplot of Fig. 2, the safety condition is violated for $\theta = 0.01$ and 0.001. These unsafe cases are strongly correlated with the observed increase in computation time. Specifically, Fig. 2 shows a sharp rise in computation time immediately preceding the point at which $h(x_k)$ becomes negative. This suggests that delayed computation can cause control inputs u_k to be applied longer than intended, thereby compromising safety. These findings highlight one consequence of increased computation time resulting from changes in relative degree. Another consequence is provided in Section 5.2.

We analyze how the sampling period influences the accuracy of the discretization approximation and, consequently, the continuous-time safety of system (22). The system is discretized with a truncation order of $N = 1$, and the controller parameters are set to $m = N_h = 4$. The constraints imposed in (20) are $\Omega \in [-0.5, 0.5]^3$ and $u \in [-0.1, 0.1]^3$. The Lipschitz constant of the output function (23) is taken as $L_h = 1$. Based on Assumption

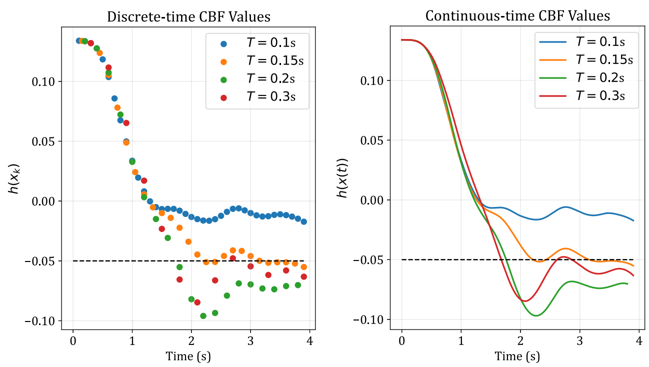

Fig. 3. Comparison of discrete-time and continuous-time CBF values under different sampling periods.

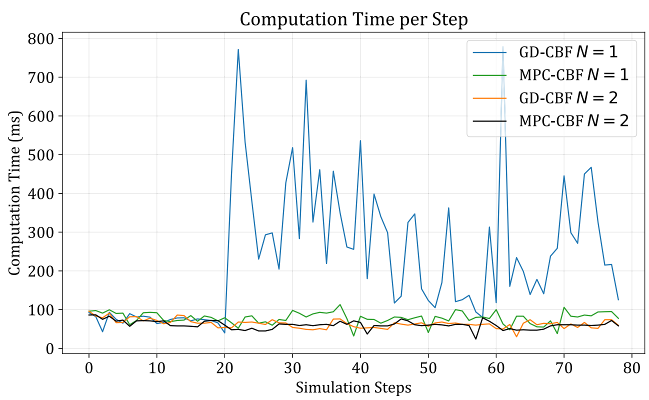

Fig. 4. Computation time comparison between the GD-CBF controller and the MPC-CBF with truncation orders $N = 1$ and $N = 2$.

1, we compute $f_{\max} = 2.64$, and $\mathbb{L}_f$ is estimated numerically as $\mathbb{L}_f \approx 1.73$. With $\delta = 0.05$, Theorem 3 yields a sampling period bound of $T \leq 0.135$. That is, $h(x_k) \geq -\delta$ holds for all $k \in \mathbb{N}$ when $T \leq 0.135$. To assess this bound, we evaluate system performance under four sampling periods: $T = 0.1$, 0.15, 0.2, and 0.3. As shown in the left subplot of Fig. 3, the condition $h(x_k) \geq -\delta$ is satisfied only for $T = 0.1$, while it is violated for the larger sampling periods. These results are in strong agreement with the theoretical prediction of Theorem 3, thereby confirming its practical relevance. Furthermore, as depicted in the right subplot of Fig. 3, the continuous-time output $h(x(t))$ performs comparably to its discrete-time counterpart $h(x_k)$ in terms of safety. This observation is consistent with the conclusion of Theorem 4.

As discussed in Remark 9, the MPC-CBF method proposed in Zeng et al. (2021) encounters difficulties when dealing with systems exhibiting high relative degrees. To illustrate this limitation, we adjust the parameters in controller (20) by setting $N_h = m = 4$ and $N = 1$. Under these settings, the relative degree of the ASDS becomes $\lceil \frac{r_c}{N} \rceil = 2$. As a result, in the MPC-CBF framework, $\Delta h(x_k)$ becomes independent of u_k, which leads to a substantial increase in computation time. Fig. 4 presents the computation times for both GD-CBF and MPC-CBF when $N = 1$. The proposed GD-CBF consistently demonstrates significantly lower computation times compared to MPC-CBF in scenarios involving high relative degrees. When the truncation order is increased from $N = 1$ to $N = 2$, the relative degree is reduced to $\lceil \frac{r_c}{N} \rceil = 1$. In this case, the MPC-CBF achieves computation times comparable to those of the GD-CBF, as shown in Fig. 4. These results highlight the superior adaptability of the proposed GD-CBF method, especially in high relative degree scenarios.

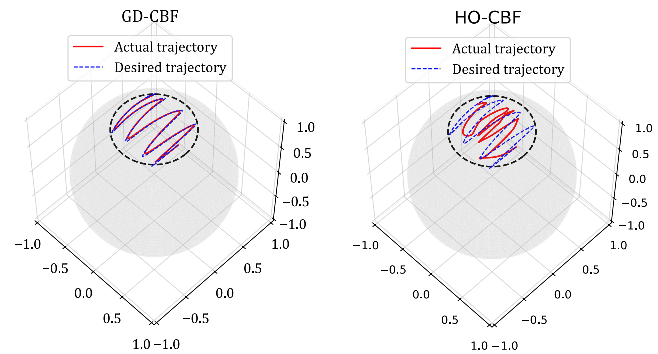

Fig. 5. Trajectories of the quadrotor body's z-axis under GD-CBF (left) and HO-CBF (right) controllers.

Our GD-CBF controller is also compared with the high-order control barrier function (HO-CBF) controller proposed in Xiao and Belta (2021). The nominal controller for HO-CBF and the corresponding parameter settings follow those in Khan and Chatterjee (2020). For our GD-CBF, we set $N_h = m = 4$ and $T = 0.05$; all other settings remain consistent with prior validations. The results are illustrated in Fig. 5. From the perspective of safety, GD-CBF performs comparably to HO-CBF in ensuring the safety of the original continuous-time system. Moreover, since our method operates on an ASDS, it achieves superior tracking performance via multi-step prediction, whereas the HO-CBF controller exhibits poor trajectory tracking performance.

5.2. Variable relative degree simulation

Beyond the challenge of increased computational delays presented in Section 5.1, variable relative degrees can also lead to safety violations due to sudden increases in control input. This subsection provides an in-depth analysis of this phenomenon.

Consider an original continuous-time system

$$\begin{cases} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -dx_1 - x_2 \\ x_1^3 \end{bmatrix} + \begin{bmatrix} 0 \\ x_2 \end{bmatrix} u, \\ h(x) = qx_2^2 - x_1 - 1, \end{cases} \tag{24}$$

where the state is $x = [x_1, x_2]^\top \in \mathbb{R}^2$, and d is a system parameter set to $d = 0.4$ in our simulations. The parameter q is chosen as $q = -0.1$ or $q = 0.1$ to represent the convex and nonconvex safe set cases, respectively. According to Jankovic (2018), states for which $x_2 = 0$ exhibit a relative degree of 2, while all other states have a relative degree of 1. The set of states with a relative degree of 2 is depicted as a black dashed line in Fig. 6.

Next, the system (24) is time-discretized with a truncation order $N = 1$ to obtain the corresponding ASDS (3). As a result, the ASDS exhibits the same relative degree as the original system (24). The ASDS is tasked with approaching a destination at $x = [0, 0]^T$, which lies outside the safe set. However, its states must remain within the safe set (denoted by the gray area in Fig. 6). Therefore, under this constraint, the optimal strategy is to converge to a state closest to the destination while staying within the safe set.

The parameters for controllers are set as $N_h = 5$, $T = 0.05$, $Q = 0.05 \cdot I$, $Q_f = I$, $R = I$ and $\Delta_u = 1$. The extended class $\mathcal{K}$ functions in (6) are all set as $\alpha_l(\cdot) = 0.9\mathcal{I}(\cdot)$, $l = 0, \ldots, m-1$. In this simulation, we fix the control input execution time to eliminate the influence of computational delay on the results. This setup allows us to isolate and analyze the observed phenomena without interference from the timing effects discussed in Section 5.1.

Under the current setting, we have $\lceil \frac{\max_{x \in \mathcal{X}} \mathcal{R}(x)}{N} \rceil = 2$. We evaluate the system's behavior under two configurations of the

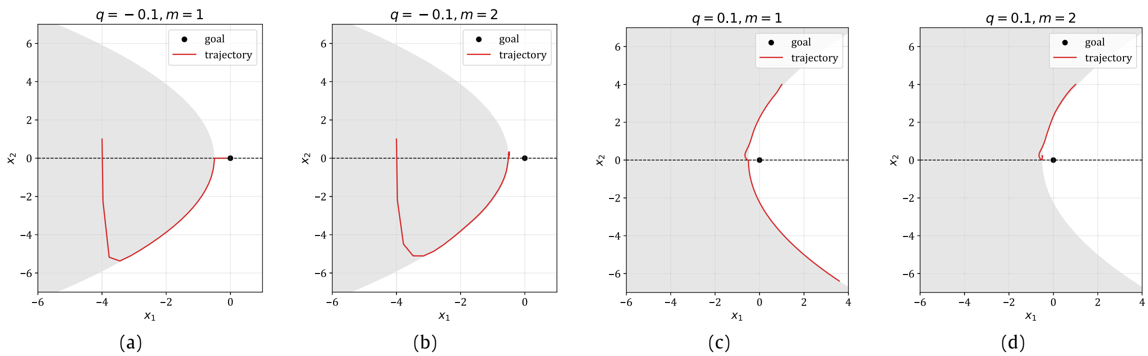

Fig. 6. Trajectories for different safety predictive horizons m under two cases: (a) $m = 1$ and (b) $m = 2$ when $q = -0.1$, and (c) $m = 1$ and (d) $m = 2$ when $q = 0.1$. The black dashed line denotes the set of states with relative degree 2, and the gray region indicates the safe set.

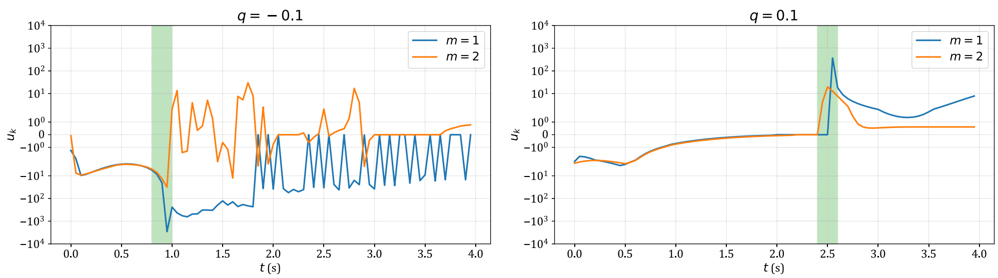

Fig. 7. Control inputs for the two cases $q = -0.1$ (left) and $q = 0.1$ (right). In both cases, the configuration with $m = 1$ exhibits a pronounced peak in the control input.

safety predictive horizon: $m = 1$ and $m = 2$, considering both convex and nonconvex safe set cases.

For the convex safe set case, i.e., $q = -0.1$, the results are shown in Fig. 6(a) and (b). When the safety predictive horizon m is set below the required threshold, i.e., $m < \lceil \frac{\max_{x \in \mathcal{X}} \mathcal{R}(x)}{N} \rceil = 2$, the trajectory may violate the safe set due to passing through states with relative degree 2. In contrast, when $m \geq \lceil \frac{\max_{x \in \mathcal{X}} \mathcal{R}(x)}{N} \rceil$, i.e., $m \geq 2$, the ASDS remains safe and converges to a stable state. These results are consistent with the theoretical guarantees provided by Theorem 2. Moreover, the observed safety violation can also be attributed to a sudden surge in the control input. As shown in the left subplot of Fig. 7, the configuration with $m = 1$ results in a control input whose absolute value exceeds 2500. This extreme value arises because $\phi_m(\cdot)$ is independent of u_k at that state. Notably, this is a distinct effect of the variable relative degree—separate from the timing-related issues discussed in Section 5.1.

In addition to the convex safe set case, we also consider a nonconvex safe set case (with $q = 0.1$). In this scenario, the trajectory is again designed to pass through the state corresponding to a relative degree of 2. Notably, when the trajectory approaches the state $[-0.5, 0]^\top$, the controller with $m = 1$ generates control inputs that are several orders of magnitude larger than those produced with $m = 2$, as highlighted by the green-shaded region in the right subplot of Fig. 7. This discrepancy leads to different trajectory behaviors, as shown in Fig. 6(c) and (d). Unlike the convex safe set case discussed earlier, however, the large control input in this nonconvex case does not result in unsafe behavior. Overall, violating the condition in Theorem 2 (i) can induce excessively large control inputs in both convex and nonconvex safe set scenarios, and such violations may, in some cases, lead to unsafe system behavior.

In conclusion, across all case studies, the proposed GD-CBF controller can ensure the safety of ASDSs by satisfying explicit conditions on the safety predictive horizon and truncation order, regardless of variations in relative degrees.

6. Conclusion

This paper has introduced a novel method termed the GD-CBF to ensure system safety. It has established a condition to ensure the discrete-time safety of the ASDS regardless of whether its relative degree is constant or state-dependent. We also ensure that the safety performance of the original system does not degrade significantly compared to that of the ASDS. Simulation results have demonstrated that our proposed GD-CBF controller can ensure the safety of the ASDS, even when relative degrees vary. We further derive an allowable range on the sampling period to ensure that the safety performance of the original continuous-time system is largely preserved when implemented via the ASDS. However, the GD-CBF controller can become computationally intensive as the safety predictive horizon and truncation order increase, which poses challenges for real-time applicability—an issue we aim to address in future work.

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